# HumanCOIN: The Human-Signal Dataset at the Scale of Egocentric Manipulation


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*https://koorye.github.io/HumanCOIN-dev*

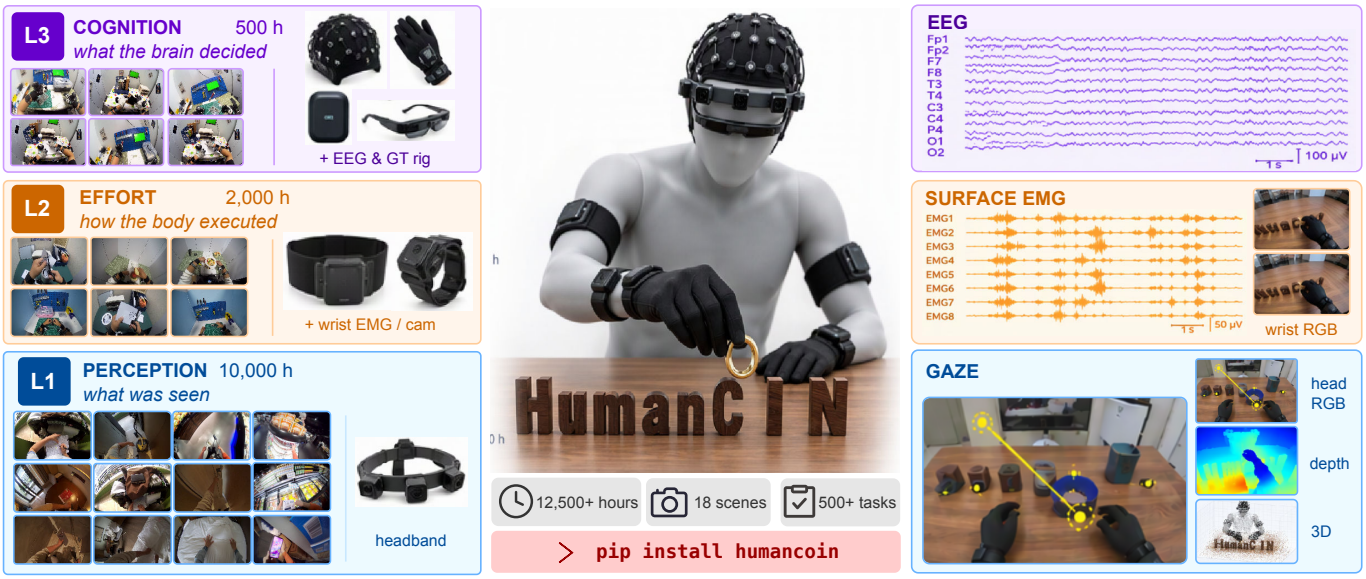



Fig. 1: Overview. Left: the three tiers, each with its hours, the question it answers, the sensors it adds, and sample views: cognition (500 h), effort (2,000 h), and perception (10,000 h). Centre: the full sensor set worn at L3, with the corpus name assembled from task objects on the desk; the corpus holds 12,500 hours across 18 scenes and more than 500 tasks. Right: what the instruments yield, namely head RGB with depth, the reconstructed 3D scene, gaze, surface EMG, and EEG.


***Abstract*—Egocentric manipulation data is becoming a central paradigm in embodied intelligence, providing paired visual–action trajectories from cameras and hand poses. These data, however, remain at a surface-level understanding of action and carry almost no record of the internal signals that reflect the operator's progression from attention to motion to thought. To close this gap, we present HumanCOIN, a large-scale egocentric manipulation dataset centred on human signals, comprising 12,500 hours of data across 18 scene classes and 500+ task entries. The dataset is organized by signal depth into three tiers: the perception tier captures egocentric vision and infers where the operator's attention lands; the effort tier adds surface EMG to measure how the body executes; and the cognition tier adds EEG to record what the nervous system decides before motion begins. Wrist views, finger kinematics, six-DoF head and wrist poses, and fine-grained textual annotations are additionally recorded to build a precise physical understanding of the manipulation process. HumanCOIN makes three lines of study possible at a scale and in a form that did not previously exist: predicting intent before an action occurs, reading the physical course of motion from muscle activity, and using brain signals to characterize more complex thought. HumanCOIN is released openly so that the field can move from imitating actions to understanding them. Our dataset, data browser, and out-of-the-box client are available at https://koorye.github.io/HumanCOIN-dev.**

## I. Introduction

Egocentric data has become the main route from human video to embodied learning. A head-mounted camera records the scene from the wearer's viewpoint, hand pose recovers what the hands did, and together they yield a visual–action trajectory: *what was seen and how the hand moved*. The recipe has scaled from passive video to instrumented manipulation [1, 2, 3, 4], and from single-scene to multi-scene corpora [1, 5]. The resulting data has enabled imitation learning [6, 7] and action recognition [1, 2, 3, 4].

However, these trajectories record the surface of manipulation, not its cause. Consider a three-second "pick up the cup": the video shows a reach and a grasp, and nothing else. Yet the act unfolds in stages the frame cannot show. The movement is planned before it begins. Attention settles on the rim before the hand arrives. The motor command reaches the forearm and scales grip force to the expected weight before the cup leaves the table. An action can therefore be imitated from the outside while the process that produced it stays unavailable: the trajectory teaches the shape of a movement without exposing the attention that aimed it, the muscle that loaded it, or the decision that committed to it.

To close that gap, HumanCOIN is built around *human signals* and organized into three tiers (Fig. 2a), comprising 12,500 hours of egocentric manipulation across 18 scene classes and 500+ task entries. The corpus follows a perception–effort–cognition axis: from the external scene the operator faces, to the peripheral muscle that executes the action, to the central activity that produces and monitors it. Signal depth increases along this axis while coverage narrows, and each tier inherits every modality of the tier below.

- **L1, Perception (10,000 h).** Records what the operator saw: multiview egocentric RGB, with attention inferred as gaze.
- **L2, Effort (2,000 h).** Records how the body executed the decision: surface EMG, whose envelope carries load and grip force before the object moves.
- **L3, Cognition (500 h).** Records what the nervous system decided before motion began: EEG, which reflects the higher-level signals of thought, such as error-related potentials that arise before any observable movement.

Beyond the tiers, the dataset provides a set of spatial modalities shared across layers: depth maps and 3D point clouds reconstructed from the egocentric views, 6-DoF head and wrist pose with finger kinematics, and step-level textual annotations with action and object descriptions. An episode is therefore not a video of hands but a queryable physical scene: where the operator stood, what they looked at, and with what posture they contacted which object are all measurable in one coordinate frame, characterizing how a manipulation unfolds. The streams are not resampled onto a common grid: each instrument keeps its own clock and native rate, capture records only the per-stream arrival offsets, and a shared spatiotemporal reference places every sample on one timeline.

We position L3 as the gold standard for these modalities: gaze, head and wrist pose, and finger kinematics are measured directly at L3 with dedicated instruments, while their counterparts at L1 and L2 are predicted from multiview egocentric video, aligning the same modality across depths in semantics, coordinate frame, and time base. This pairing turns L3 into a supervised reference for the shallow tiers: a predictor run on L1 or L2 video can be scored frame by frame against what L3 actually measured, and the residual decomposed by scene, task, and depth.

Relative to existing corpora, HumanCOIN differs less in scale than in coverage: three depths under one task set and one time base. L1 supplies scale and scene diversity, L2 moves the operation from appearance to effort, and L3 provides verifiable targets measured with dedicated instruments. These human signals capture the generative process behind each video–action pair, making the causal chain from intention to execution available for modeling rather than only for observation, closing the gap left by first-person corpora that record outcomes alone.

All streams are released in their native form, with the per-session calibration and clock offsets, so that users can re-synchronise, re-calibrate, or re-annotate the corpus without returning to the raw recordings.

The HumanCOIN is released under an open-source license, encouraging the community to build upon and extend the dataset. In summary, our contributions are threefold:

- **A large-scale human-signal dataset.** The corpus holds 12,500 hours over 18 scenes and 500+ tasks, organized into three tiers with strict inheritance (Section III).
- **A synchronized acquisition paradigm.** Human signals and first-person manipulation are recorded under one clock at three instrumentation depths (Section IV).
- **An open ecosystem for the community.** Acquisition software, a data browser and downloader, and an out-of-the-box client are released with the corpus (Section IV-C).

## II. Related Work

### A. Egocentric Vision Corpora

Egocentric video has been scaled along two axes, hours and annotation density. Ego4D [1] collected 3,670 hours from 931 participants in nine countries and defined benchmarks for episodic memory, forecasting, and social interaction. EPIC-KITCHENS [2] traded breadth for density, annotating cooking with verb–noun segments; its hundred-hour extension, EPIC-KITCHENS-100 [5], added 89K action segments. EGTEA Gaze+ [8] recorded eye movements over 28 hours of cooking and turned attention into a label. Ego-Exo4D [9] added a synchronized third-person view: 1,286 hours from 740 participants, annotated with skills, keysteps, and proficiency.

Assembly101 [3] instrumented tabletop assembly with 4,321 multi-view videos; ARCTIC [4] captured bimanual manipulation of articulated objects with 2.1M frames and 3D hand–object annotations; HOI4D [10] recorded 2.4M RGB-D frames of category-level hand–object interaction; and HoloAssist [11] annotated 166 hours of human–assistant collaboration, including mistakes and their corrections.

A newer generation raises the bar on geometry and duration: Nymeria [12] captures multimodal daily motion with full-body and hand tracking, HD-EPIC [13] adds 3D object and gaze grounding to kitchen annotation, EgoDex [14] records 829 hours of manipulation with on-device 3D hand tracking, and EgoLife [15] follows six participants for a week. All remain external observations: none records muscle or cortex during the act. EgoSuite-Open100K [16] has opened the first 10,000 hours of a planned 100,000-hour programme of egocentric manipulation, with 3D hand and body pose over 10,000 tasks and 500 environments, and Xperience-10M [17] releases 10,000 hours and 2.88B RGB frames of in-the-wild human experience with depth, camera pose, and captions. Both record the outside of the action at a scale no research corpus reaches. Neither records gaze, muscle, or cortex, and neither pairs the same task with a shallower version of itself.

These corpora answer what happened, and how the hands acted. However, they do not record the signals that produced the action: attention, muscle, and brain are all missing. HumanCOIN adds them, and organizes them by depth so that the same task is observed at several physiological levels.

### B. Physiological and Neural Sensing

Surface EMG has long served as the interface between motor intent and machine. Farina et al. [18] review how motor-unit activity is extracted from the surface signal and used to decode intended movement and force, and Ninapro [19] standardised forearm recordings across subjects. EEG-based interfaces have decoded motor preparation, error monitoring, and cognitive load for decades [20]. Gaze provides the earliest behavioural signature of intent: it reaches the target of a pointing movement before the hand does [21], and appearance-based trackers [22] made the signal recordable outside the laboratory; in everyday tasks fixations run ahead of the hand and mark the objects about to be acted on [23, 24]; and in object manipulation gaze reaches the contact point before the fingers close [25]. Pupil diameter adds a physiological readout of processing load [26], and when gaze and EEG are recorded together, fixation-related potentials allow cortical activity to be analysed time-locked to what is being looked at [27].

The neuroergonomics programme [28] argued that these measurements belong inside real tasks. Most studies nevertheless remain laboratory experiments: the task is designed to elicit a signal, scenes are controlled, participants few and trials short, so behaviour, muscle and cortex are rarely recorded on one clock. The gap has begun to close from the neural side: EgoBrain [29] pairs a 32-channel EEG headset with egocentric video over 61 hours from 40 participants and releases both, the first corpus to put cortex on the same clock as first-person video. It is seated and scripted, however, over 29 daily actions in one setting, with no gaze, no muscle, and no scene geometry. What remains missing is corpus-scale recording of several such signals at once, in realistic scenes, with the same task observed at several physiological depths.

### C. Human Data for Robot Learning

R3M [30] pretrains a visual representation on Ego4D and transfers it to manipulation; VRB [31] extracts affordances from human videos as a representation for robot policies; EgoMimic [32] scales imitation learning from egocentric human video captured with a portable rig. Earlier systems (MIME [33], BC-Z [34], DexMV [35], RoboAgent [36], and TCN [37]) and offline-demonstration studies [38] established the recipe that these corpora now scale. At corpus scale, Open X-Embodiment [6] pools demonstrations across robots, DROID [39] collects in the wild, and ALOHA [40] shows what low-cost bimanual hardware yields; RT-2 [41] and Diffusion Policy [42] train on such corpora. These methods inherit appearance and affordance from the human, but not the signals that produced the action.

A newer line adds modalities to the policy itself. Any2Policy [43] trains a modality-agnostic encoder that accepts text, audio, images, or point clouds in any combination and connects them to a policy head. Chain-of-Modality [44] prompts a vision-language model to chain modalities from human demonstration videos (images, armband muscle activity, and audio) into task plans with explicit control parameters. Play to the Score [45] fuses multiple sensory streams with stage-guided dynamic weighting during manipulation.

These works show that extra modalities help a policy, but none is built on a corpus that records them from the human at several depths; HumanCOIN supplies one, and Table I sets that gap out row by row.

## III. Data Distribution

HumanCOIN holds 12,500 hours of egocentric manipulation, split 10,000 / 2,000 / 500 across L1, L2, and L3, over 18 scene classes and 500+ task entries. This section describes how the data is distributed across depth, scene, and task, and how the three tiers are related.

### A. The Depth Axis

Egocentric manipulation data answers what the world looked like while the hands changed it. It does not answer three deeper questions: what the operator perceived, how the body executed, and what the brain decided before the movement began. A single kit rarely covers all three: the deeper the layer, the costlier and more intrusive the hardware. HumanCOIN therefore tiers by signal depth (perception, effort, cognition), each tier answering its question with the smallest kit possible. Depth also shapes how the data is collected: the deeper the tier, the weaker the signal and the more it depends on clear event boundaries, so collection moves from undisturbed recording toward a controlled protocol, shrinking real-world coverage.

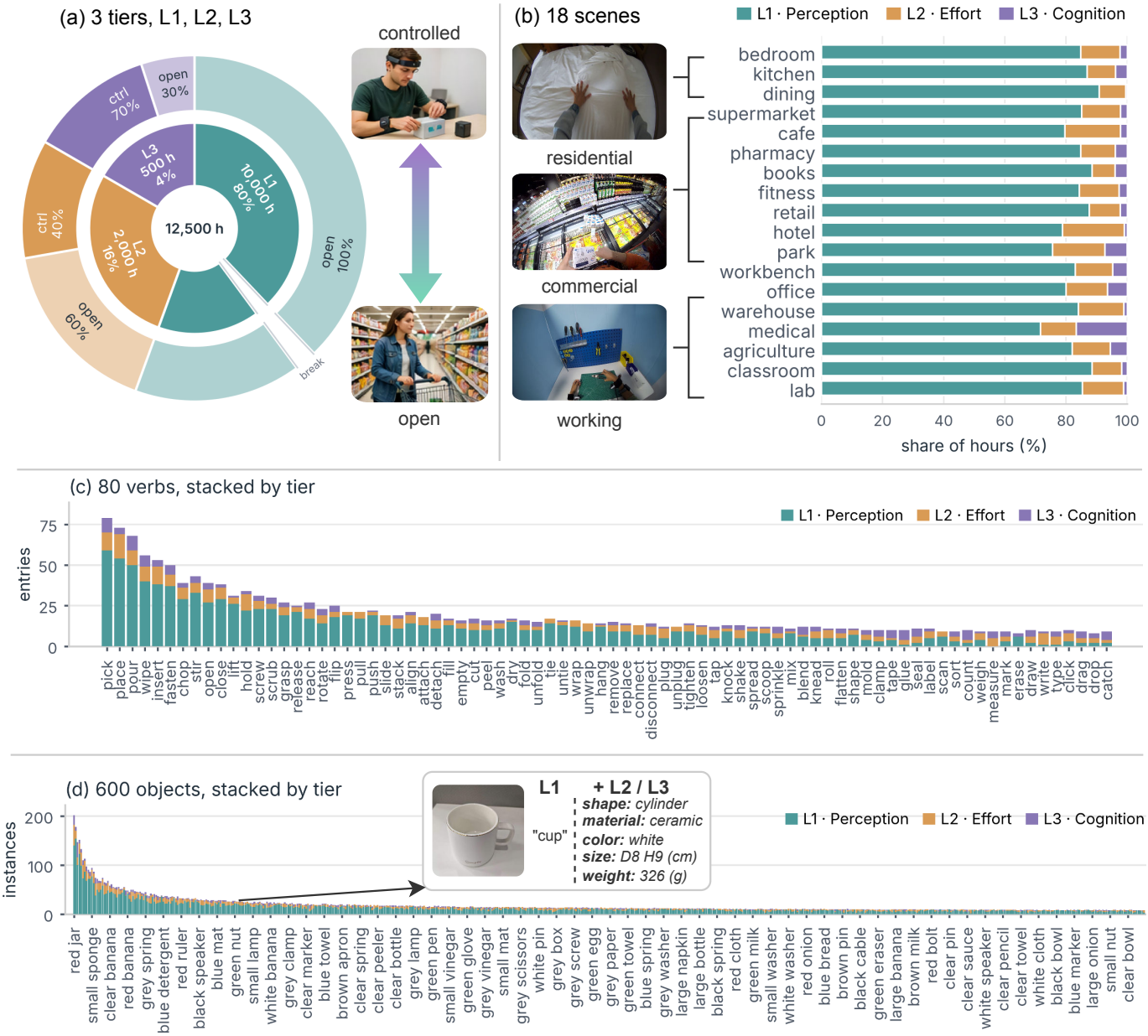



Fig. 2: HumanCOIN in one view. (a) The three tiers and how their hours split: 12,500 h in total, at 10,000 / 2,000 / 500 h (80 / 16 / 4%) for L1 / L2 / L3, with the share of each tier's hours collected in the controlled and in the open setting; the two photos contrast the settings. (b) The 18 scene classes, grouped into residential, commercial, and working zones, with the tier split of each class's hours. (c) The 80 most frequent verb types, stacked by tier: a few dominant verbs head a long tail. (d) The 600 object entries, stacked by tier; the inset shows one object as a name at L1 and as a full record at L2 and L3, with shape, material, colour, size, and weight.

### Three questions

**Q1.** What did the operator see, and how much of it can be recovered from the head alone?

**Q2.** How did the body execute the action?

**Q3.** What did the brain decide before the muscle moved?

**L1, Perception: what was seen.** This tier barely disturbs the day: no task is set and no scene is staged, the operator simply goes about their life while the headband records. Four lenses cover the left-front, right-front, left, and right views, with the acquisition computer inside the band; gaze is estimated from those views, and depth is reconstructed from the overlap of

the two front lenses. Attention and depth are therefore inferred from pixels rather than measured, and the design bet is that much of this can indeed be inferred, which is what makes L1 cheap enough to run for ten thousand hours of everyday scenes.

**L2, Effort: how the body executed.** Here tasks are set and goals given, with a task sheet that fixes object placement and step order (Section IV-B), but the scene stays everyday. The body is instrumented where the command reaches the muscle: two wrist cameras add the near-hand view, and one eight-channel armband on the palm side of each forearm records surface EMG. The data therefore carry task labels while the action stays close to real life, at the cost of a narrower scene range.

**L3, Cognition: what the brain decided.** At this depth the task has to be written as a multi-step script at a fixed station, so that boundaries such as reach, grasp, and release exist and repeat; EEG and gaze are interpretable only against such boundaries. Both the person and the scene are instrumented, so the deepest tier serves as the reference for the others: an eye tracker for gaze and pupil, three 6-DoF trackers for head and wrists, two 20-node gloves for finger joints, an 8/32/128-channel EEG cap for cortex, and a third-person RGB camera that views the operator and the desk from outside the head (Fig. 4). The cost is a scene range that shrinks to the desk.

The three tiers are not three independent datasets but a ladder of inheritance: each tier adds new modalities on top of the previous one while keeping every earlier modality. The model can therefore move from what the hand did to where the eye looked, how the muscle exerted force, and when the cortex decided, while the visual observation of the task stays comparable across tiers, the precondition for putting external behaviour and internal physiology into one coordinate frame and one timeline.

### *B. Scene Distribution*

The value of egocentric data depends largely on how far it generalizes in the real world. The scene distribution should therefore not be trimmed for convenience of collection; it should stay as close as possible to the distribution of real life. The ontology divides scenes into three zones by where people actually act, and the tiers do not cover the zones equally. The 18 classes, with one example of each, are shown in Appendix B.

**Residential.** Indoor spaces where people live day to day, characterized by ample time, little disturbance, and repeatability. Typical scenes include the bedroom, kitchen, dining room, and living room.

**Commercial.** Places of everyday consumption and service, characterized by moving people, open space, and the need for unobtrusive equipment. Typical scenes include the supermarket, café, pharmacy, bookstore, and gym.

**Working.** Spaces where people go to get work done, characterized by a structured task, a fixed station, and room to deploy instruments. Typical scenes include the workbench, office, warehouse, medical, agriculture, classroom, and lab.

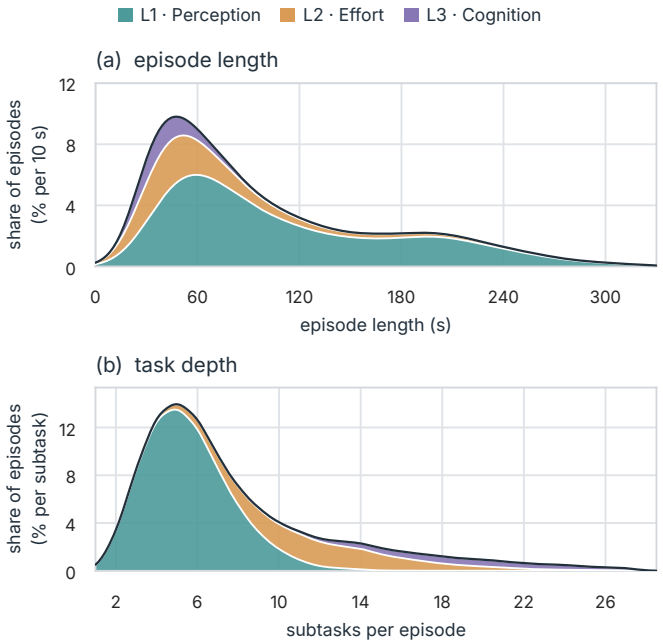



Fig. 3: Episode length and task depth by tier. (a) Continuous episode-length density, stacked by tier: each tier's kernel density is weighted by its share of episodes, so the stack is one probability density (area 100%) that peaks at 40–60 s and decays into a long tail near 5 min; the deeper tiers concentrate at shorter lengths. (b) The same stacked density for subtasks per episode; the composition shifts from few subtasks at L1 to many at L3.

### *C. Task Distribution*

A task entry is the unit of reuse: one captured segment, its ordered step sentences, and the objects it touches. The 500+ entries span cooking, assembly, cleaning, errands, and teaching, and differ in step count, object set, and manipulation difficulty; that variation is what allows the corpus to be split by domain, by horizon, or by difficulty instead of being used as one monolithic training set. We characterize these tasks along two dimensions, action and object (Fig. 2c,d):

**Action: more than a hundred verb types.** The verbs are taken from the step sentences recorded during capture and form an open set, and they differ in granularity: some describe an overall transfer, others a detail such as alignment or force. Their frequency follows a Zipf-like long tail: the ten most frequent verbs cover nearly half of the entries, while the remaining ninety-odd cover the other half, many appearing only dozens of times (Fig. 2c). The vocabulary also expands with depth: L1 is annotated with coarse, common verbs, L2 adds task-specific verbs, and L3 adds compositional ones, so the same progression that adds modalities also adds vocabulary. The complete lexicon is in Appendix C.

**Object: more than a thousand instances.** The depth of the object record follows the tiers (Fig. 2d). At L1 an object is only a name. At L2 and L3 the same object also carries an RGB colour, a shape, a size given in the parameters its shape calls for, a material, and a weight, so the record becomes ground

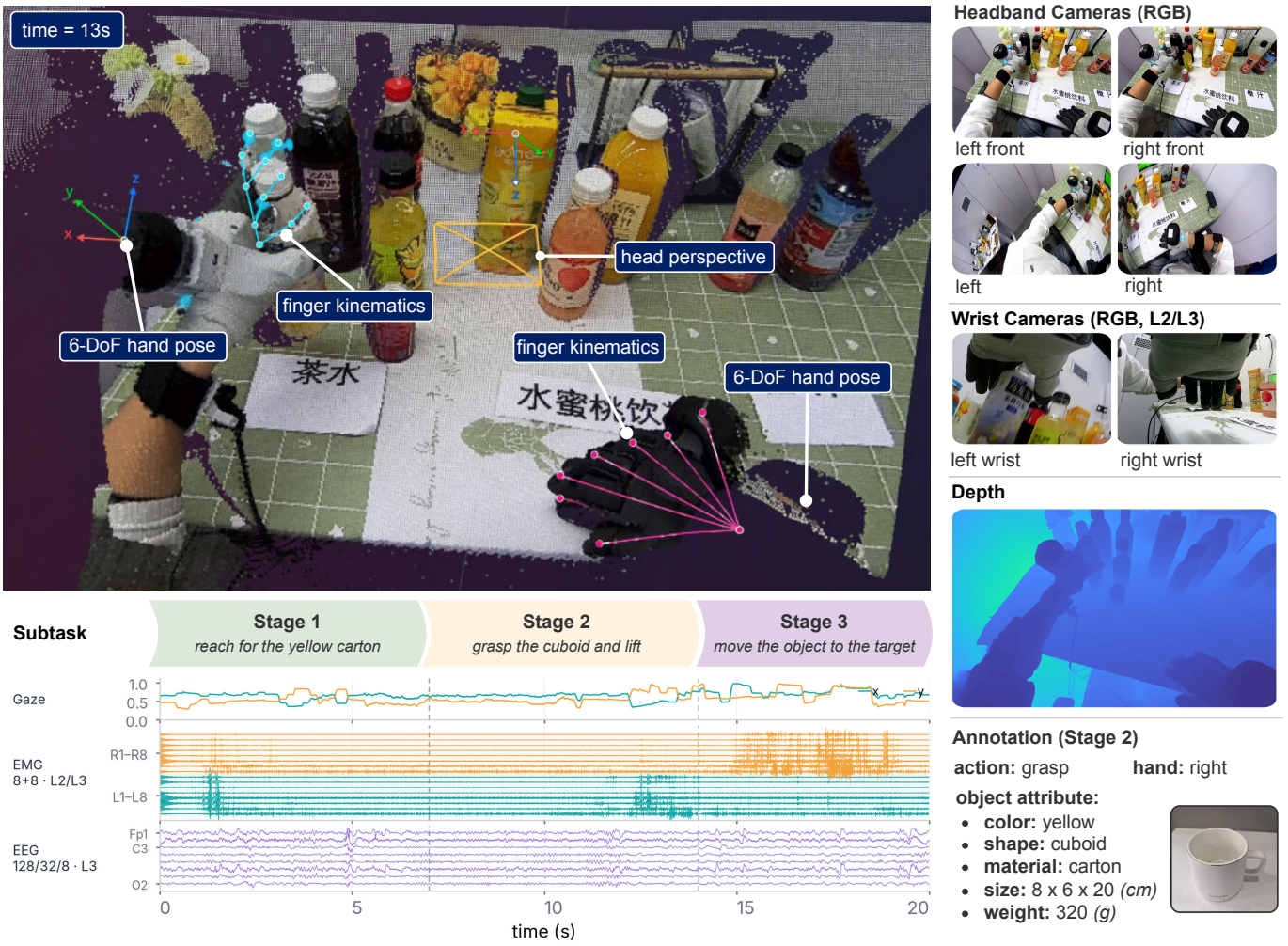



Fig. 4: One episode in full. The environment carries the point cloud, the head perspective, the 6-DoF hand pose, and the finger kinematics, all in the same coordinate system. The four head-mounted views, the two wrist views (L2/L3), and the depth map are shown once per stream. The episode is split into three stages, each with its subtask written inside the stage band; the boxed card expands the current subtask into the atomic action, the hand in use, and the object attributes (colour, shape, size, material, and weight). Below, gaze, 8+8-channel left and right wrist EMG (L2/L3), and 128/32/8-channel EEG (L3) share the same timeline.

truth rather than a label. These attributes separate tasks that look alike but are not: a full mug and an empty one load the hand differently, and a 12 g pill bottle and a 45 g box fail in different ways. The shape vocabulary and its size parameters are listed in Appendix D.

Taken together, the task distribution of HumanCOIN is a naturally long-tailed one. It preserves the diversity inherent in everyday activity: a few high-frequency actions and common objects dominate, while many low-frequency actions and long-tail objects extend to the end of the distribution. The deeper tiers, meanwhile, constrain the task through a fixed station and a script, which makes the action division finer and the object record more complete. The result is a corpus that spans from natural behaviour to complex controlled tasks, with the same task observed at several physiological depths.

### *D. Episode Duration and Task Stages*

Episode duration determines how much context and how many steps a single sample carries, and it largely constrains the kind of training and evaluation the sample can support. The duration of HumanCOIN is not set uniformly; it varies systematically with tier, task type, and difficulty.

**Duration distribution (Fig. 3a).** The distribution is unimodal with a long tail: most episodes concentrate between 40 and 60 s, decaying on both sides, with the shortest near 10 s and the longest near 5 min. The tiers differ systematically. L1 has the widest range and a larger share of long episodes; L2 and L3 shift toward the short end, and the compression at L3 is driven mainly by electrode stability rather than by the task.

**Task stages (Fig. 3b).** Subtasks per episode count the stages a task is composed of rather than the temporal granularity

TABLE I: Prior egocentric datasets against HumanCOIN. ✓ measured, P predicted from video, R reconstructed from video, ✓* on a subset only, a cross marks a modality not provided. 3D names a reconstructed depth or point-cloud stream, Fingers joint-level hand tracking, and Subtask and Object anno. step-level and object-property annotation. Frames are counted at 30 fps where the source reports no count, and n/r where it reports none; † the first 10,000 hours are open. The shaded rows are HumanCOIN: the aggregate, then its three tiers in paler type.

| Dataset | Hours | Tasks | Frames | Views | 3D | Fingers | Gaze | sEMG | EEG | Subtask anno. | Object anno. |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Ego4D [1] | 3,670 | n/a | 396M | 1 ego | ✗ | ✗ | ✓* | ✗ | ✗ | ✓ | ✗ |
| Ego-Exo4D [9] | 1,286 | n/a | 139M | 1 ego, exo | R | ✗ | ✓ | ✗ | ✗ | ✓ | ✗ |
| EPIC-KITCHENS [5] | 100 | 125 | 12M | 1 ego | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ | ✗ |
| EGTEA Gaze+ [8] | 28 | n/a | 3.0M | 1 ego | ✗ | ✗ | ✓ | ✗ | ✗ | ✓ | ✗ |
| Nymeria [12] | 1,200 | n/a | 130M | 1 ego | ✗ | ✓ | ✓ | ✗ | ✗ | ✓ | ✗ |
| EgoDex [14] | 829 | 194 | 90M | 1 ego | ✗ | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ |
| EgoSuite-Open100K [16] | ∼10,000† | 10k+ | n/r | 1 ego, 2 wrist | R | ✓ | ✗ | ✗ | ✗ | ✓ | ✗ |
| Xperience-10M [17] | 10,000 | n/a | 2.88B | 1 ego | R | ✓ | ✗ | ✗ | ✗ | ✓ | ✗ |
| EgoBrain [29] | 61 | 29 | 6.6M | 1 ego | ✗ | ✗ | ✗ | ✗ | ✓ | ✓ | ✗ |
| **HumanCOIN** | 12,500 | 500+ | 1.35B | 4 ego, 2 wrist*, exo* | R | ✓ / P | ✓ / P | ✓* | ✓* | ✓ | ✓ |
| L1, perception | 10,000 | 500+ | 1.08B | 4 ego | R | P | P | ✗ | ✗ | ✓ | ✗ |
| L2, effort | 2,000 | 500+ | 216M | 4 ego, 2 wrist | R | P | P | ✓ | ✗ | ✓ | ✓ |
| L3, cognition | 500 | 500+ | 54M | 4 ego, 2 wrist, exo | R | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

of annotation. L1 consists mainly of independent segments of natural activity, with few marked stages; L3 consists mainly of complex long-horizon tasks, with the most stages.

Overall, HumanCOIN shows clear tier differences along both duration and task structure: the shallow tiers hold short, natural, and diverse everyday segments, while the deep tiers concentrate on long-sequence, complex, and standardized tasks. Duration characterizes the recording window, stage count characterizes task structure, and together they form the corpus's spectrum from natural behaviour to complex controlled tasks.

### *E. Inside an Episode*

Fig. 4 opens one L3 episode in full: the 3D scene at the centre, the streams of the same interval around it, and the subtask timeline along the bottom. It is shown at L3 because that is the tier where the body is instrumented: the gaze, the hands, and the wrists that L1 and L2 infer from video are measured here.

**Views.** Four egocentric views pair the two front lenses (left-front and right-front) with the left and right views, and at L2 and L3 two wrist views add the hands at work. A depth map reconstructed from the front pair is stored beside them, and in the controlled setting a third-person view records the operator and the desk from outside. The frames are stored as recorded, with no annotation drawn over the pixels.

**Geometry.** The point cloud is reconstructed from the two front views; the head, both wrists, the finger joints, and the gaze are placed in the same frame, so a grasp is located in the scene rather than only in the image.

**Physiology.** EMG is drawn as traces beside the video, eight channels per arm at L2 and L3, and EEG as 8, 32, or 128 channels at L3. Both are read at the timestamps of the frames, so a muscle burst, a cortical deflection, and a hand movement line up frame by frame.

**Annotation.** Each episode is cut into stages, and each stage carries a subtask description together with the action the step names, the hand that executed it, and the objects it touched. At L2 and L3 the object record carries detailed attributes: a photo, a hex colour, a shape, a material, a weight, and a size given in the parameters its shape calls for (radius and height for a cylinder, for instance).

An episode is therefore one record aligned on a single timeline and in a single frame: the views and the point cloud give the scene, the poses, joint angles, and gaze give the body's place in it, EMG and EEG give the physiological state behind the movement, and the annotation gives each step its place in the task.

### *F. Comparison with Existing Datasets*

Table I compares these datasets on what they record rather than on scale. All of them observe the task from the head; Ego-Exo4D adds third-person views and EgoSuite wrist cameras, but every row above the rule records the visible outcome of manipulation; only EgoBrain records the cortex, and none records the muscle.

Finger-level hand pose is not ours alone either: Nymeria and EgoDex read it from head-mounted devices, and EgoSuite reports 3D hand pose for both hands, but in each it stands alone, whereas here it is predicted at L1 and L2 and measured at L3 on the same task. Gaze is the one human signal several already carry, but always on its own: nowhere is the same quantity available at two depths at once. HumanCOIN records gaze, pose, and finger kinematics at L3 and predicts the same quantities from video at L1 and L2, so an estimator can be scored against the dataset itself rather than against a separate benchmark.

The annotation columns narrow the claim. Subtask annotation is common, under names that differ—verb–noun segments, action segments, keysteps, narrations, motion

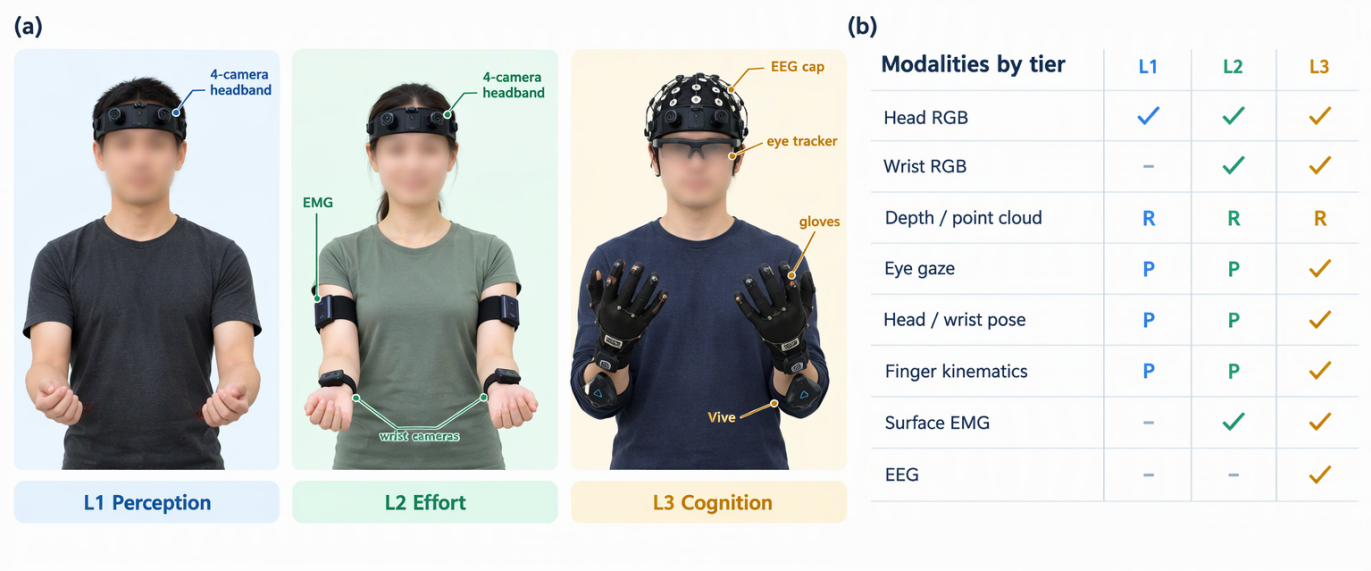



Fig. 5: Instrumentation and modalities by tier. (a) Each figure is labelled only with the devices that tier adds: L1 the four-camera headband, L2 wrist cameras and surface EMG, L3 the EEG cap with a head tracker, the eye tracker, the gloves, and the wrist trackers. (b) Modality availability, with measured (✓), predicted (P), reconstructed (R), and absent (—) entries colour-coded by tier.

segments—but object annotation is not: elsewhere an object is only a name, while at L2 and L3 each instance is measured for shape, material, size, and weight, which is what lets an EMG envelope be read against a load nobody had to estimate. The result is a corpus that spans from natural behaviour to complex controlled tasks, with the same task observed at several physiological depths.

## IV. Hardware Configuration

The signals summarized above are produced by independent instruments, each keeping its own clock, its own native sampling rate, and its own coordinate frame. This section describes what is worn at each tier, the two settings in which the kit is deployed, the two-phase protocol that organizes a controlled session, and how the resulting streams are stored.

### A. The Instrument Set

The four-camera headband is in-house. Four lenses cover the left-front, right-front, left, and right views of the scene; at L2 and L3 two wrist cameras add the near-hand view, mounted on each forearm and aimed along the hand. The band also carries the acquisition computer: it records every stream in the open setting and forwards them over a single link to the bench host in the controlled setting. The kits are strictly nested, and each tier adds one class of instrument to the tier below:

- **L1, Perception**: the four-camera headband only, recording head RGB.
- **L2, Effort**: adds two wrist cameras and bilateral surface EMG, eight channels per forearm, at the motor command's last peripheral station.
- **L3, Cognition**: adds a wearable binocular eye tracker with its scene camera, three 6-DoF trackers on the head and both wrists, a pair of 20-node gloves, an EEG cap with 8, 32, or 128 channels, and a third-person RGB view that records the operator and the desk from outside the head.

One host receives every stream and stamps arrival offsets; no instrument is daisy-chained through another, and in the open setting that host is the headband computer. Phase and run transitions are also driven through a parallel-port box into the EEG amplifier, so protocol events are stamped on the amplifier's clock as well as on the host's. All cameras, the headband, and the eye tracker are calibrated: distortion is estimated once, every image is stored undistorted, and the calibration file ships with the data. Poses share one coordinate frame, defined once and reused downstream: the desk runs along $-y$, left is $+x$, and up is $+z$, with millimetres for translation and degrees for joint angles.

Fig. 5(a) shows the three tiers with only the devices each tier adds, and Fig. 5(b) turns the same information into a matrix. The matrix separates what is measured from what is inferred. Head RGB, wrist RGB, third-person RGB, surface EMG, and EEG are measured at the tiers that carry them. Gaze, head/wrist pose, and finger kinematics are predicted from egocentric video at L1 and L2 and measured at L3, which is why the corpus can be used both to train estimators on the shallow tiers and to score them against the gold standard. Depth and point clouds are reconstructed from the egocentric RGB at every tier. Models, channel counts, sampling rates, and interfaces are listed in Appendix E.

### B. Settings and Protocol

Two things decide how a tier is collected: how much of the everyday scene it must preserve, and how much signal fidelity




Fig. 6: The two cued phases of a controlled session (e.g., *pick up the cup*). (a) Imagery: the screen turns blue and the operator, facing the screen, sits still and rehearses the task; only EEG is retained. (b) Execution: the screen turns green and the same operator manipulates the objects on the desk.

its hardware demands. The two pull in opposite directions, and the corpus resolves them tier by tier.

L1 and L2 want the scene more than signal depth. Every instrument they add is small enough to be worn and connects to a headband-mounted computer, so the operator moves through everyday environments (standing, turning, walking, and gesturing) with no bench and no cable in the way. This is the *open* setting: L1 and L2 are collected in it throughout, and it also carries L3, trading EEG density for scene coverage.

L3 is where the pull reverses. A 32- or 128-channel cap needs a stable posture: electrode contact and impedance degrade with movement, and the denser the cap, the less the record tolerates it. The eye tracker, the gloves, and the wrist trackers load the same platform, so acquisition wants a bench that does not move. Sessions run in short blocks, with a re-baseline between blocks to limit drift. L3 is collected under both settings:

- **Controlled setting.** At a dedicated desk, the amplifier and the host stay on the bench and the operator holds still; the cap is 32 or 128 channels, and the session runs imagery then execution.
- **Open setting.** In everyday scenes such as a supermarket, a café, or a kitchen, the kit runs on the headband computer; L3 uses the 8-channel cap, and the session runs execution only.

The controlled setting also carries the paradigm that makes the EEG informative, built around the high-density cap. Preparation and motor imagery are brief, and the movement that follows can bury them, so the session separates rehearsal from performance (Fig. 6):

- **Imagine (phase 1).** The screen turns blue: the operator faces it, sits still, and rehearses the task sequence without moving, so that motor imagery and preparatory EEG are recorded without movement artefacts; only EEG is retained.
- **Execute (phase 2).** The screen turns green: the same operator manipulates the objects on the desk, and every modality is recorded.

Each controlled episode therefore has three segments, not just the two cues: preparation while the operator settles, imagery of the task sequence, and execution of that sequence. Because the first two are still, the EEG records the task as it is planned rather than as it is performed: the step sequence and the action each step will require, with no movement to mask it. The high-density cap exists for that pairing: the same task as a plan, recorded while the body is still, and as a performance seconds later. Open settings (L1, L2, and L3 with the 8-channel cap) record the execution segment alone.

Collection follows a fixed procedure. L1 is recorded without guidance: the operator is told only to go about the activity. L2 and L3 use task sheets: the operator arranges the objects as drawn and follows the step order marked on the sheet, so that the same task is staged the same way across sessions. At L3, tasks are drawn in random order rather than blocked, which keeps the operator engaged and keeps the EEG from drifting into drowsiness. Operators also rotate between scenes at a fixed interval, so that no one works a single scene for a whole shift and no scene is shaped by a single person's habits.

All operators are trained to one standard before collection: how to wear each device, how to run a session, and how to record it. L3 adds two requirements: operators are trained to keep the head still during acquisition and are responsible for seating, gelling, and maintaining the EEG cap so that impedance stays within range; participants assigned to L3 are also screened so that the EEG is clear and stable. Every hour of collection is followed by fifteen minutes of rest, to keep the operator's attention and energy up. Appendix A reports the operators, sessions, and per-operator hours behind these totals.

### C. Data Storage

Throughout, instruments are *joined by a reference, not fused into an enclosure*. Each keeps its own clock, its own native

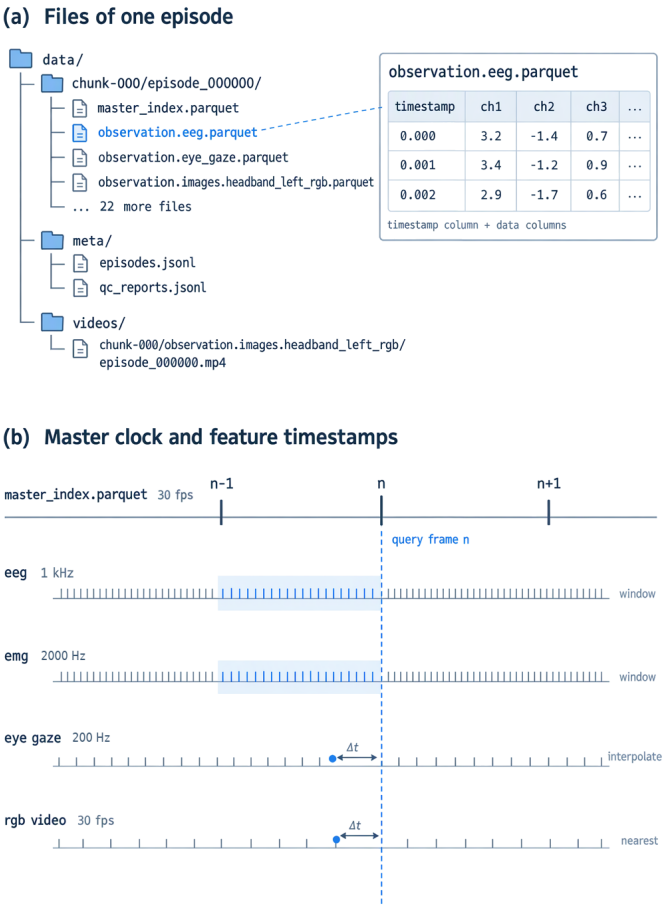



Fig. 7: One episode on disk, and how it is read. (a) One Parquet file per stream, each a timestamp column plus data columns, beside a master index. (b) Timestamps are relative to the start of the recording, so no feature sits on a fixed grid; a query fixes a master rate and resolves each feature from its own timestamps (window, nearest, interpolation).

sampling rate, and its own coordinate frame; forcing them onto a single timeline would discard the timing the corpus exists to keep. HumanCOIN therefore stores every modality as it was recorded, with no resampling and no alignment at write time: synchronization is a property the data *declares*, not one the pipeline *bakes in*. The corpus is written as many asynchronous files, each at its own rate.

At write time, every episode keeps one Parquet table per modality (Fig. 7a), one file per observation stream: the camera views, EEG, gaze, hand pose, glove skeleton, three device poses, head and wrist IMUs, and wrist EMG. `videos/` holds one encoded video per camera view, and `meta/` carries the calibration, the quality reports, and the episode metadata. Every sample is stamped relative to the start of the recording, and that difference is stored as given, not snapped to a master frame, so non-uniform intervals and device jitter survive into the file. The complete directory layout is in Appendix F.

Reading is fully decoupled from writing. A separate master-index file is sampled at a fixed rate, and the reader chooses that rate (30 fps by default); a master frame then indexes every modality through its own timestamps, resolved by nearest sample, interpolation, or all samples in a window such as $(-1/\mathrm{FPS}, 0]$ (Fig. 7b).

Quality is enforced before writing rather than after. Every recording is checked for dropped modalities, black or frozen frames, missing channels, and segments where the modality set is incomplete; a segment that fails is removed from the release instead of being stored with a flag. The surviving episodes then pass independent manual reviews for each modality, and the verdicts are stored in the metadata.

The format is a modification of LeRobot [46] and remains compatible with it. Installing `pip install humancoin` is enough to preview, download, and use the data; the acquisition software, the hardware kit, and the documentation are released alongside it[1].

## V. Data Processing

Processing runs per stream first and meets a shared contract only at the end (Fig. 8). Five groups cover the pipeline: synchronization and calibration, geometry and motion, gaze, electrophysiology, and annotation. Each modality is handled by a module that knows its physics, and the modules meet only when every stream has been reduced to timestamped samples in a declared frame.

### A. Synchronization and Calibration

We use soft synchronization. The recorder stamps every sample relative to the start of the recording, and the sync host records arrival offsets for the streams that share a transport; the two together place each sample on one timeline without resampling. Calibration (intrinsics and distortion for every camera, the tracker-to-frame transform, and the eye-tracker calibration) ships with the episode, so a reader can re-derive the geometry instead of inheriting a pre-warped one. Because neither offsets nor calibrations are applied at write time, a reader can align at any custom rate and choose nearest sample, interpolation, or a window (Section IV-C). Residuals between the chosen rate and the samples actually read are written to the per-episode alignment report, so a downstream analysis can audit the timing it inherited.

### B. Geometry and Motion

Depth is built from RGB. The left-front and right-front lenses are hardware-timestamp synchronized, so every stereo pair comes from the same instant; the pair, together with the camera parameters, is passed to FoundationStereo [47], which returns a depth map stored with the intrinsics that produced it. Point clouds are formed per frame by projecting depth through those intrinsics and are subsampled to a fixed budget.

Motion follows the same route. At L3 three 6-DoF trackers give the head and both wrists directly (120 Hz, millimetres, one shared frame), and the gloves give 20 joint angles per hand (200 Hz, degrees). At L1 and L2 hand position and pose are estimated in RGB with HaMeR [48], followed by an additional smoothing pass, and then projected into 3D through

[1] https://github.com/Koorye/Multi-Frequency-LeRobot

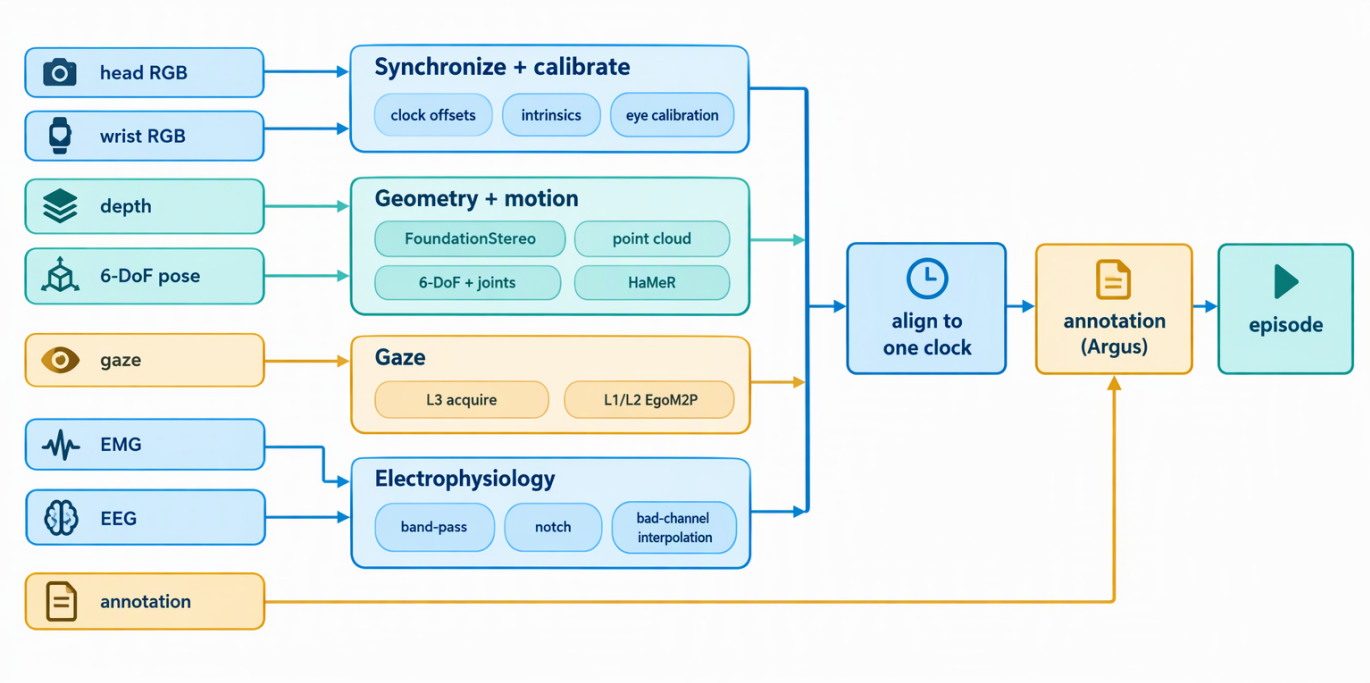



Fig. 8: Processing pipeline. Each stream enters the group that handles its physics: synchronization and calibration (clock offsets, camera intrinsics, eye calibration); geometry and motion (stereo depth from FoundationStereo, point cloud, 6-DoF poses with finger joints, HaMeR hand estimates); gaze (acquisition at L3, EgoM2P at L1 and L2); and electrophysiology (band-pass filtering, notch, bad-channel interpolation). The groups then align to one clock and pass through annotation before an episode is written.

the depth map (Appendix G compares this choice against alternatives). All poses (head, both hands, hand kinematics) and the environment point cloud are mapped into one unified coordinate frame.

### C. Gaze

Gaze is obtained by acquisition and by estimation. At L3 the eye tracker is calibrated per participant and per session, and its 200 Hz output, the $x, y$ pixel of gaze in the eye-camera image, is stored with the calibration that produced it. At L1 and L2 gaze comes from EgoM2P [49], a model that takes an ego RGB sequence and returns head view and trajectory, eye movement, and depth; we keep its eye-movement output. On L3 episodes, where the eye tracker supplies a reference, the predicted gaze departs from the measured one by roughly 7° in median angular error (Appendix H, which also shows the trails frame by frame and reports per-scene errors).

### D. Electrophysiology

The electrophysiology is filtered per modality before it is stored, and each stage is recorded so that a reader can undo it. EMG is band-passed between 20 and 450 Hz with a 50 Hz comb notch, rectified, and reduced to an envelope per channel, with the raw trace kept beside it.

EEG passes through four stages. A 0.5–40 Hz band-pass and a 50 Hz notch remove drift and line noise while keeping the delta-to-beta range in which motor-imagery and error-related potentials live. Bad channels are flagged on three criteria—impedance above the session threshold, decorrelation from the neighbouring channels, and a variance that departs from the montage by a robust $z$-score—and repaired by spherical-spline interpolation over the retained ones. Ocular artefacts are removed by ICA, with components selected by frontal topography and by their correlation with the blink and saccade signal of the eye tracker recorded at L3; muscle and movement artefacts are attenuated by artifact subspace reconstruction; intervals in which the head IMU marks a turn are labelled rather than cut. The cleaned trace is finally re-referenced to the average of the retained channels. Raw trace, cleaned trace, and the mask that records every changed channel, component, and interval are released together, so a reader can work on the uncleaned signal or on any prefix of the chain.

Motion cannot be removed from a recording in which the operator moves, and even the deepest tier is stationary only by protocol (Section IV-B). The cleaned EEG is therefore signal plus residual motion noise: we release the metrics that bound it—impedance pass rate, share of interpolated channels, and share of samples marked artefactual, reported per setting—and leave the choice of a stricter rejection threshold to the user. EMG is recorded at 2000 Hz in millivolts, eight channels per arm; EEG at 1000 Hz over 128 saline channels or at 256 Hz over 32 or 8 gel channels, in microvolts, with every channel named by brain region and its impedance logged in kilohms as a quality input.

### E. Annotation and Segmentation

Every episode is annotated at the level of task steps (Fig. 4): temporally bounded segments carrying the step sentence, the action it names, the hand in use, and every object the sentence reaches. Annotation is produced with Argus [50], an open-source harness for dense annotation of robot-learning data, running on top of Qwen3-VL-32B [51]. Argus ingests an episode in its native format (here our LeRobot-derived layout) and reasons jointly over video, state, and sensor evidence to produce dense labels: action phases, subgoals, key events, outcomes, operator mistakes and recoveries, and object-state changes, each with an execution and outcome score and with the evidence and confidence behind it. It exports the labels as JSON and as LeRobot language columns, and a human reviewer checks every export before it enters the release. Objects are not mere name strings: for L2 and L3 every object used was measured for size and weight, and the measurement is linked to the object instances appearing in each trajectory. Collective references ("the cups") are expanded to their member objects, and hand-in-use is read from wrist-tracker displacement inside the step window when no explicit label exists.

L1 needs a different route. Because passive collection does not intervene, a raw L1 recording is one continuous stream that lasts hours, and episodes are cut after the fact: frames and the available annotation are passed to a VLM, which proposes episode boundaries and step segments, and a human reviewer confirms or adjusts every boundary before release. This is why L1 episodes vary widely in length while controlled L3 sessions are scripted, and why an L1 boundary carries the uncertainty of a segmentation rather than the certainty of a cue.

## VI. Discussion

HumanCOIN is a concept corpus: its value is measured not by the size of any single tier but by the comparisons its structure makes possible. Three families of questions motivated the design, and each is answerable by the same comparative move: hold the task ontology fixed, vary signal depth across tiers, and let each physiological channel earn its place only where vision cannot decide on its own.

### A. Limitations

HumanCOIN is built to provide a structure that can be validated rather than to substitute for that validation. This paper concentrates on the acquisition paradigm, the tier design, and the release; downstream policy experiments, baselines for anticipation and error detection, and a cross-site evaluation of the quality gate fall outside its scope and are left for the community to build on. The L1 and L2 estimators of gaze, hand pose, and finger kinematics ship with the data, and their systematic evaluation is likewise left for later work. The value of the tier structure is therefore supported by design at this stage; the size and conditions of any predictive gain remain to be characterized by experiments that use the corpus.

The second limitation concerns coverage. L3 accounts for four percent of the hours and is concentrated in desk scenes, so findings about the deepest tier are best read as properties of a deliberately focused sample rather than of manipulation at large. The controlled protocol is shaped by electrode stability, which sets a practical bound on how long the densest episodes can run, and the scene and task mix, though designed, is not a random sample of manipulation in the world. A third limitation is annotation: step boundaries and object attributes are human judgements, and L1 boundaries in particular come from an automatic segmentation, so they carry a boundary uncertainty that the scripted L3 sessions do not.

### B. Research Directions

The corpus supports three lines of exploration, and all three run on the same comparative move: hold the task fixed across tiers, vary signal depth, and let each channel earn its place where vision cannot decide.

- **Prediction.** How much of the future (which object will be grasped, when a reach will begin, whether an attempt will be abandoned) is readable from gaze and cortex before the hand moves, and at what lead time.
- **Physical properties.** Two objects can look identical and load the hand differently; EMG makes weight and grip effort readable from the operator, with object attributes as ground truth.
- **Failure and recovery.** Failures are rare and their interesting events precede any visible change; the failure-rich L3 tasks put error-related cortex, gaze to the fault, and the correction on one timeline.

Together the three directions test one premise from three sides, and a negative result on any of them would be as informative as a positive one.

## VII. Conclusion

HumanCOIN reframes what an egocentric manipulation corpus can be. Instead of recording only what the world looked like while hands changed it, it records the same task at three signal depths (perception, effort, and cognition), so that what the operator saw, how the body executed, and what the cortex decided before either are measured on one timeline. The tiers inherit strictly, share one task ontology, and differ in how they are collected: passive and open-ended at L1, active but open at L2, and active and controlled at L3. The result is a corpus that spans from natural behaviour to complex controlled tasks, with the same task observed at several physiological depths. The corpus is released openly for research use, and the acquisition software, the hardware kit, and the documentation are released alongside it.

## VIII. Data Statement

All participants were informed of the purpose and procedure, gave written consent before any recording, and could withdraw at any time without penalty. Episodes are released only in de-identified form: no personally identifying information is included, faces are blurred in every released view, and L1 and L2 collection follows the same protocol; EEG

and EMG are physiological data released under the same de-identification. Code is released under Apache-2.0 and data under CC BY-NC 4.0.

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# APPENDIX

## A. PARTICIPANTS AND COLLECTION

The corpus was collected by 120 operators: all 120 worked at L1, 48 of them also at L2, and 24 at L3, so the same task is observed at all three depths by the same hands. Ages 20–45, 58 women, 11 left-handed; all adults, recruited through a professional agency and trained to one standard before collection (Section IV-B). L3 candidates are screened for signal quality, and of 32 candidates 24 entered collection.

Collection ran over 2,940 sessions at 6 sites, spread across the 18 scene classes. An operator contributed a median of 83 hours (interquartile range 46–118; range 12–210), and a session lasted a median of 240 minutes (range 90–480). L1 sessions are long and unscripted; L2 and L3 sessions are shorter, bounded by electrode stability and by attention. Operators rotate scenes at a fixed interval and rest fifteen minutes for every hour of collection (Section IV-B), so no scene is shaped by a single operator's habits. All participants gave written consent before recording (Section VIII).

TABLE II: Collection by tier. Hours are corpus totals; Hours/op. is the median over the operators who worked at that tier, and Hours/day the mean over their active collection days.

| Tier | Operators | Sessions | Hours | Hours/op. | Hours/day |
|---|---|---|---|---|---|
| L1 | 120 | 2,040 | 10,000 | 83 | 5.0 |
| L2 | 48 | 640 | 2,000 | 42 | 4.2 |
| L3 | 24 | 260 | 500 | 21 | 3.0 |

## B. SCENE CLASSES

The corpus spans 18 scene classes, grouped into residential (bedroom, kitchen, dining), commercial (supermarket, cafe, pharmacy, books, fitness, retail, hotel), and working zones (park, workbench, office, warehouse, medical, agriculture, classroom, lab). Fig. 9 shows one impression of each class; the release ships recorded footage for every class, and the per-class hour split is in Fig. 2b.

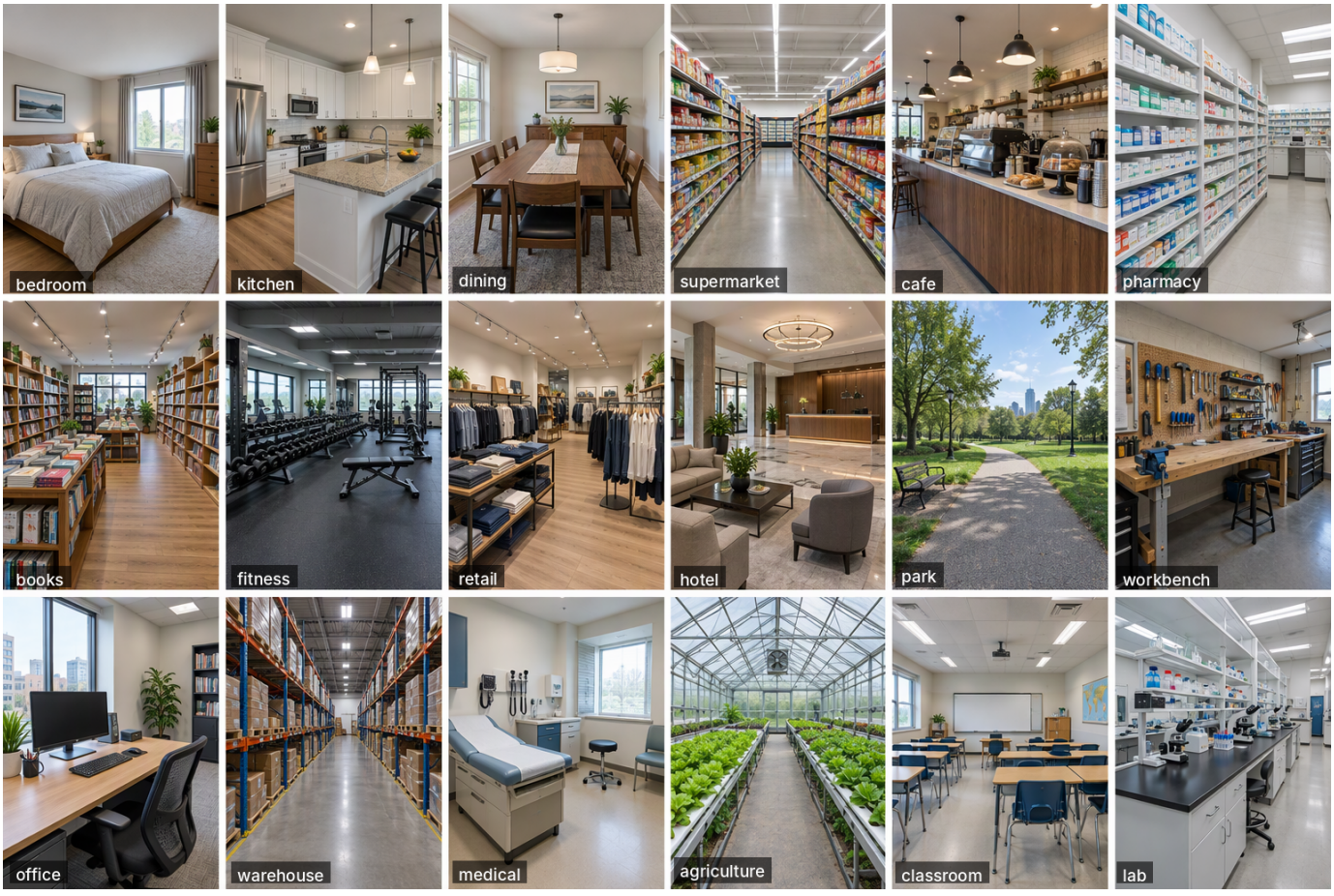



Fig. 9: The 18 scene classes, labelled left to right and top to bottom. The tiles are illustrative renderings of the classes, not frames from the corpus; recorded footage for each class ships with the release.

## C. Verb Lexicon

The annotation uses an open verb vocabulary taken from the capture's step sentences. Table III lists the most frequent verbs with a one-line gloss and the tiers in which each occurs; the full lexicon, with counts per tier, ships with the release.

TABLE III: Verb lexicon. Glosses state the action boundary used by the annotators; tiers give where the verb is observed.

| Verb | Gloss | Tiers |
|---|---|---|
| pick | grasp an object and lift it clear of its support | L1–L3 |
| place | put an object down at a target location | L1–L3 |
| lift | raise an object without moving it laterally | L1–L3 |
| lower | move an object downward under control | L1–L3 |
| carry | transport an object while holding it | L1–L3 |
| move | displace an object within the workspace | L1–L3 |
| push | displace an object by force applied away from the body | L1–L3 |
| pull | displace an object by force applied toward the body | L1–L3 |
| drag | move an object while it stays in contact with the surface | L2–L3 |
| slide | translate an object along a surface | L2–L3 |
| rotate | turn an object about a vertical axis | L2–L3 |
| flip | turn an object over | L2–L3 |
| hand over | pass an object to another person | L1–L3 |
| insert | place a part into a receptacle and seat it | L2–L3 |
| align | bring two parts into a shared axis or pose | L2–L3 |
| seat | press a part fully into its receptacle | L3 |
| screw | drive a threaded fastener with a twisting motion | L3 |
| unscrew | release a threaded fastener | L3 |
| fasten | fix two parts together | L3 |
| tighten | increase the force holding two parts together | L3 |
| loosen | reduce the force holding two parts together | L3 |
| connect | join two components so they function together | L3 |
| plug | insert a connector into its socket | L3 |
| attach | fix an accessory onto a host object | L3 |
| detach | remove an accessory from its host | L3 |
| assemble | build a whole from several parts | L3 |
| disassemble | take a whole apart into its parts | L3 |
| chop | cut food into pieces with a blade | L2–L3 |
| slice | cut a thin piece from a larger body | L2–L3 |
| stir | agitate a mixture in a vessel | L1–L3 |
| mix | combine two or more substances | L1–L3 |
| pour | tip a vessel so its contents fall into a target | L1–L3 |
| scoop | take up material with a concave tool | L2–L3 |
| spread | distribute a substance over a surface | L2–L3 |
| fold | bend a flexible object back onto itself | L2–L3 |
| knead | work a soft mass by repeated pressure | L3 |
| peel | strip an outer layer from an object | L2–L3 |
| season | add a condiment to food | L2–L3 |
| wipe | clean a surface with a cloth in contact | L1–L3 |
| scrub | clean a surface with repeated pressure | L2–L3 |
| sweep | gather debris across a floor | L2–L3 |
| mop | clean a floor with a wet tool | L2–L3 |
| rinse | remove residue with water | L2–L3 |
| spray | apply a liquid as a fine mist | L2–L3 |
| dry | remove moisture from an object or surface | L2–L3 |
| polish | rub a surface to a shine | L3 |
| check | inspect an object against an expected state | L1–L3 |
| compare | place two objects or states side by side | L2–L3 |
| verify | confirm a state against a specification | L3 |
| count | enumerate a set of objects | L2–L3 |
| sort | separate objects into categories | L2–L3 |
| stack | place objects one on another | L1–L3 |
| open | move a closure to its open state | L1–L3 |
| close | move a closure to its closed state | L1–L3 |
| seal | close a container so its contents cannot escape | L2–L3 |
| label | attach a marker carrying information | L2–L3 |
| pack | place objects into a container for transport | L2–L3 |

### D. Object Shapes

Every object instance is described by its name, colour, shape, size, material, and weight. Size is recorded as the parameters of its shape rather than as a single number, so the same schema serves a mug, a box, and a sheet. Fig. 10 gives the shapes in use, their parameters, and their share of the object instances; lengths are measured with a caliper to $\pm 0.1$ mm and weights on a scale to $\pm 1$ g.

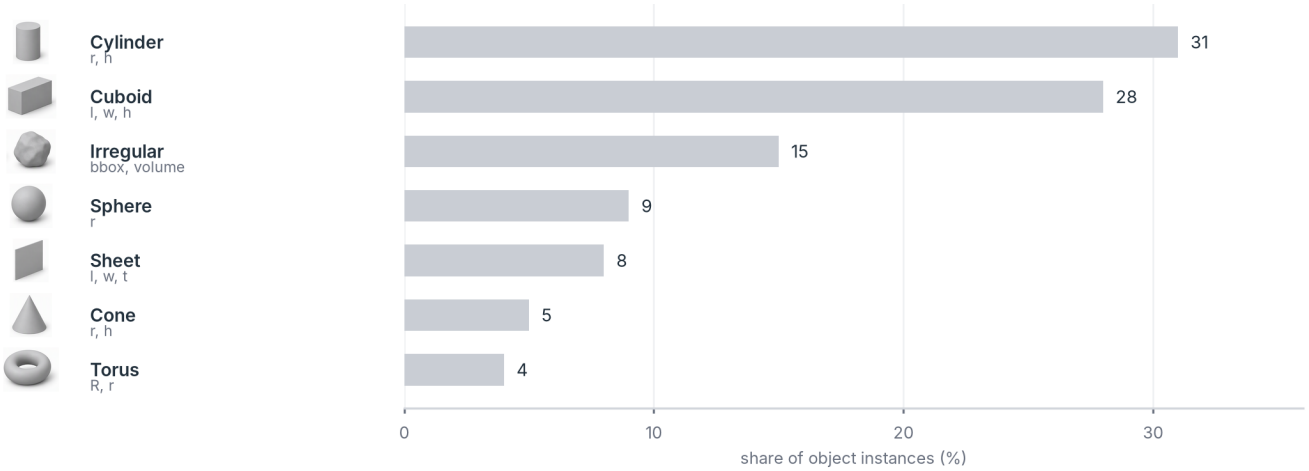



Fig. 10: Object shapes, their size parameters, and their share of instances. A parameter marked $r$ is a radius, $h$ a height, and $l, w, t$ length, width, and thickness. The grey solids are schematic renders of each shape class.

### E. Hardware Bill of Materials

TABLE IV: Stream specifications. All cameras record RGB, are undistorted, and run at 30 fps; the third-person view exists only in the controlled setting.

| Stream | Device (model) | Resolution / channels | Rate | Unit |
|---|---|---|---|---|
| Head RGB $\times 4$ | four-camera headband, Ego Headband V1.5 (in-house) | $1920 \times 1080$ each; left-front, right-front, left, right; clock hardware-synced | 30 fps | RGB |
| Wrist RGB $\times 2$ | JIERUIWEITONG FH868 | $640 \times 480$ | 30 fps | RGB |
| Eye camera | Pupil Labs Neon | $1920 \times 1080$ | 30 fps | RGB |
| Third-person view | Intel RealSense D455f | $640 \times 480$ | 30 fps | RGB |
| Gaze | Pupil Labs Neon | $x, y$ pixel in the eye-camera image | 200 Hz | px |
| Head / wrist pose $\times 3$ | HTC VIVE Tracker 3.0 | one shared frame: $-y$ desk, $+x$ left, $+z$ up | 120 Hz | mm |
| Finger kinematics | MANUS Metaglove Pro Haptics | 20 joint angles per hand | 200 Hz | deg |
| Surface EMG | WLEMGUV2A armbands $\times 2$ | 8 channels per arm | 2000 Hz | mV |
| EEG (saline) | Neuroscan SynAmps2 (Curry) | 128 channels, brain-region labels, impedance logged | 1000 Hz | $\mu$V |
| EEG (gel) | BrainCo BCIGo | 32 or 8 channels, brain-region labels, impedance logged | 256 Hz | $\mu$V |
| Sync marker | UDP marker | — | 30 Hz | — |

Table IV lists the instruments used in collection, with the resolution, channel count, rate, and unit of every stream. The headband is in-house; the remaining devices are commercial and are used as delivered.

**Headband.** Four synchronised lenses sit on a rigid frame with a fixed baseline, so the stereo pair used for depth keeps its geometry across sessions; an inertial unit and the acquisition computer are mounted behind them, and the battery is carried at the back of the band. The band is fitted to the head with a replaceable inner liner, which also keeps the lenses at a repeatable height and angle.

**Wrist kit.** Each wrist carries one camera on a strap aimed along the hand and one armband with eight dry electrodes on the palm side; the two are cabled together so the forearm does not pull on the camera. The kit is worn over the glove at L3 and directly on the forearm at L2.

**Cognition kit.** The L3 kit adds a binocular eye tracker with its scene camera, three 6-DoF trackers (head and both wrists), a pair of 20-node gloves, and an EEG cap. Two cap types are used: a 128-channel saline cap for the controlled setting and a 32- or 8-channel gel cap for the open setting; both are seated and checked for impedance before each block.

**Synchronisation hardware.** A UDP marker is emitted by the recorder, and phase and run transitions are additionally driven through a parallel-port box into the EEG amplifier, so protocol events are stamped on the amplifier clock as well as on the host clock. Cameras, headband, and eye tracker are calibrated before each session against a printed target; the resulting intrinsics, distortion, and tracker transforms ship with the episode.

**Consumables and spares.** Saline, gel, syringes, electrode paste, spare straps, liners, and batteries are restocked per shift; a failed cap or camera is swapped and the session restarted rather than repaired in place, so that a released episode never mixes two calibrations.

## F. Episode File Layout

Every episode is self-contained and follows a chunked hierarchy compatible with a modified LeRobot v3.1 release. The directory is named after the session (collector, date, clock); the example below holds 1,444 master frames (48.1 s), eight videos, one audio track, and 26 Parquet tables, 493 MB in total (data 21 MB, video 470 MB, audio 1.5 MB, meta 200 KB).

```
L3-003-2026-10-21-18-13-11/
  data/chunk-000/episode_000000/
    master_index.parquet
    action.parquet
    observation.<stream>.parquet
  videos/chunk-000/<stream>/episode_000000.mp4
  audios/chunk-000/<stream>/episode_000000.wav
  meta/
    info.json  episodes.jsonl  tasks.jsonl
    collect_info.jsonl  calibrations.jsonl
    alignment_check.jsonl  qc_reports.jsonl
    episodes_stats.jsonl
```

The 26 tables are `action`, `master_index`, eight `observation.images.*` timestamp tables, and sixteen `observation.*` tables: `eeg`, `eye_gaze`, `eye_imu`, `hand_pose`, `hand_skeleton`, `head_pose`, `headband_imu0`, `headband_microphone`, `left_wrist_emg`, `left_wrist_imu`, `left_wrist_pose`, `marker`, `right_wrist_emg`, `right_wrist_imu`, `right_wrist_pose`, and `state`.

TABLE V: Representative tables in one L3 episode (1,444 frames, 48.1 s). Rows are per-sample; every table also carries `timestamp` and `episode_index`.

| Table | Rows | Rate | Content |
|---|---|---|---|
| master_index | 1,444 | 30 Hz | clock, frame and task index |
| action, state | 26,017 | 542 Hz | 58-D pose and joints |
| eeg | 12,031 | 250 Hz | 32 ch + IO, TRIG |
| emg (L/R) | 96,163 | 2 kHz | 8 ch per arm |
| eye_gaze | 9,608 | 200 Hz | $x, y$ in eye image |
| hand_pose | 26,017 | 542 Hz | 20 angles per hand |
| hand_skeleton | 26,017 | 542 Hz | 50 nodes $\times$ 6 |
| head/wrist_pose | 3,564 | 74 Hz | $x, y, z$, roll, pitch, yaw |
| images.* | 1,396–1,442 | 30 Hz | frame timestamps only |

**Signals.** Each stream is one Parquet file at its native rate, with a `timestamp` column in seconds from episode start and an `episode_index`; columns are named after channels (`FP1...`), joints (`node00_x...`), or `dim_0...` for EMG, so no stream is resampled. Pixels are not stored in Parquet: `observation.images.*.parquet` lists the timestamp of every frame and the frames themselves live in the matching mp4 (H.264, 30 fps). Audio is 16 kHz mono PCM in wav, with its Parquet file recording the sample count per master frame.

**Metadata.** `info.json` records the codebase version, master feature, fps, path templates, chunk size, and the shape, rate, and tolerance of every feature. `episodes.jsonl` and `tasks.jsonl` map indices to the task string and episode length. `collect_info.jsonl` stores collector, hardware, scene, the object list (colour, shape, dimensions, material, pose), and free-text notes. `calibrations.jsonl` stores the camera chain (intrinsics, distortion, and extrinsics between the four lenses), the camera–IMU chain, and the IMU parameters. `alignment_check.jsonl` reports, per feature, the tolerance, number of readings, violations, and maximum offset; `qc_reports.jsonl` reports per-stream rate, gaps, empty windows, and a pass/warn level; `episodes_stats.jsonl` stores per-feature min, max, mean, and standard deviation.

**Reading.** The reader fixes a master rate (30 fps by default) and walks `master_index`; each row gives a time, and every other stream is resolved from its own timestamps by nearest sample, linear interpolation, or all samples in a window such as $(-1/\mathrm{FPS}, 0]$. Nothing is stored on the master grid, so a stream can be re-read at any rate without degradation: a query that

needs raw EMG reads the 2 kHz table directly, while a query that needs one frame per 33 ms reads the index and never touches the other tables.

## G. Comparing Hand-Pose Strategies

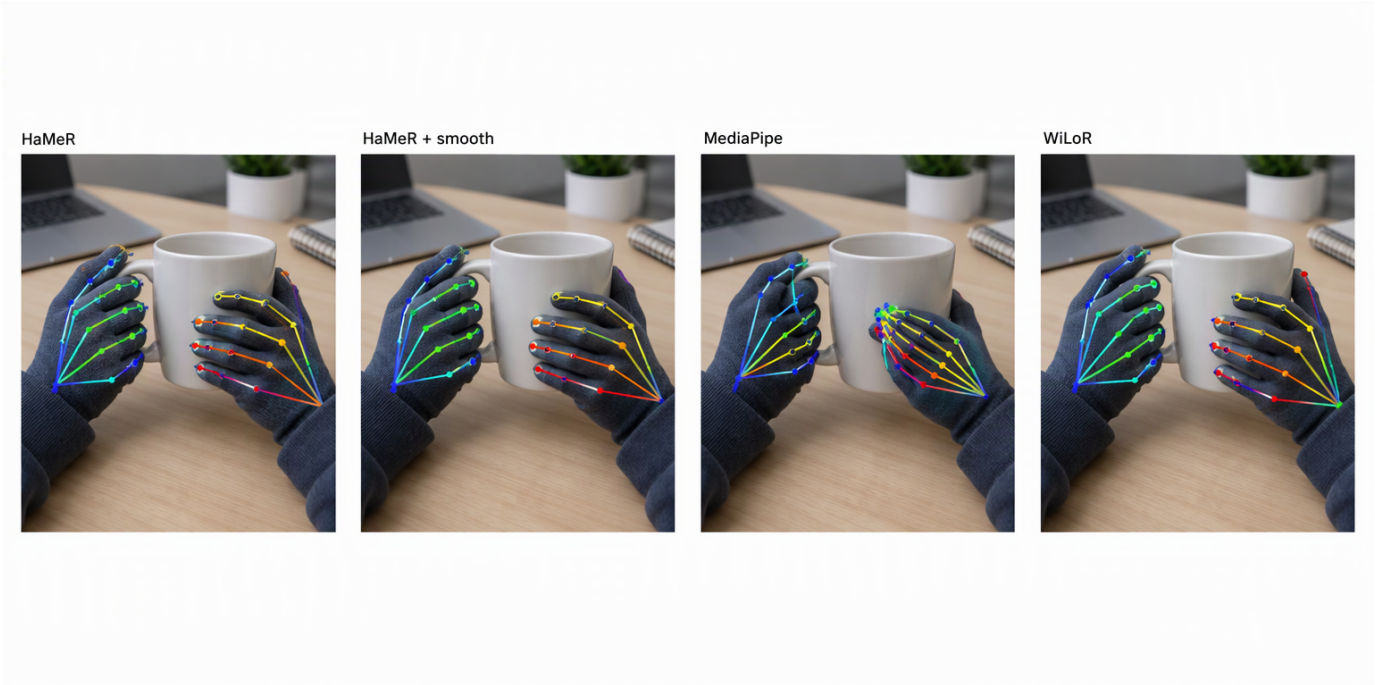



Fig. 11: Hand pose under four strategies on the same frame: HaMeR, HaMeR with the smoothing pass used in the release, MediaPipe, and WiLoR. MediaPipe collapses the fingers during contact; WiLoR drifts at the fingertips; smoothing removes the frame-to-frame jitter of HaMeR without changing its fit.

TABLE VI: Hand-pose strategies on the same L3 frames. Jitter is the frame-to-frame motion of the wrist keypoint; dropout is the share of frames with no estimate; the last column is the dominant failure we observed.

| Strategy | Jitter (mm) | Dropout (%) | Main failure |
|---|---|---|---|
| HaMeR | 2.4 | 3.1 | fingertip jitter |
| HaMeR + smooth | 0.9 | 3.1 | — |
| MediaPipe | 4.8 | 9.7 | finger collapse |
| WiLoR | 3.6 | 6.2 | fingertip drift |

**Methods.** HaMeR [48] regresses MANO hand parameters from a single RGB crop with a transformer and is the base of the released L1/L2 tracks. MediaPipe [52] fits a lightweight 21-landmark model in real time. WiLoR [53] extends transformer-based regression to in-the-wild hands with a MANO head trained on larger corpora.

HaMeR has no ground-truth reference at L1 and L2, so we compare strategies rather than measuring absolute error. On the same episodes we ran HaMeR, HaMeR with the temporal smoothing used in the release, MediaPipe [52], and WiLoR [53], and inspected their skeletons on frames with contact, occlusion, and fast motion (Fig. 11). HaMeR with smoothing gives the most stable wrist and finger tracks; MediaPipe collapses fingers during contact, and WiLoR drifts at the fingertips under occlusion. The release therefore uses HaMeR with the smoothing pass.

In practice the smoothing pass costs one frame of latency and removes most of the visible jitter, so the release ships it as the default for L1 and L2 hand tracks; the unsmoothed tracks remain available for users who prefer to filter themselves.

## H. Comparing Gaze Strategies

**Estimators.** EgoM2P [49] is a feed-forward model that takes an egocentric RGB clip and returns head trajectory, eye movement, and depth in one pass; we use only its eye-movement output. GLC [54] is a global–local transformer for egocentric gaze: it correlates a global scene token with local eye-region evidence and predicts a gaze direction in the camera frame. It was trained on EGTEA and Ego4D and is the closest published estimator to the L1/L2 setting.

**Protocol.** Both models see the same frames of the same L3 episodes. GLC returns a direction, which we map into the eye-camera plane with the headband intrinsics stored in `calibrations.jsonl`; EgoM2P returns image coordinates directly.

We score both against the eye tracker with median and 90th-percentile angular error, frame-to-frame jitter, and dropout, so that the comparison isolates the model rather than the evaluation.

Because L3 records gaze with the eye tracker, it doubles as a test bed for the estimator used at L1 and L2. We ran EgoM2P on the ego RGB of L3 episodes and compared its output with the tracker record over the same interval. The predicted trajectory follows the measured one in overall shape, but lags at saccades and drifts during rapid head turns; the median angular error is about 7°, with the largest deviations at scene changes and fast motion (Fig. 12). This is the error a user should expect when training on L1 or L2 gaze.

TABLE VII: Gaze sources on L3, by scene. Median and 90th-percentile angular error in degrees; jitter is the frame-to-frame change of the predicted gaze direction. The eye tracker is the reference (GT) and has zero error by definition.

| Scene | Median (°) | $p_{90}$ (°) | Jitter (°/frame) |
|---|---|---|---|
| Kitchen | 6.8 | 14.2 | 1.1 |
| Desk assembly | 7.4 | 15.6 | 1.3 |
| Supermarket | 8.1 | 17.9 | 1.6 |
| Overall | 7.0 | 15.4 | 1.2 |
| Eye tracker (GT) | 0.0 | 0.0 | 0.0 |

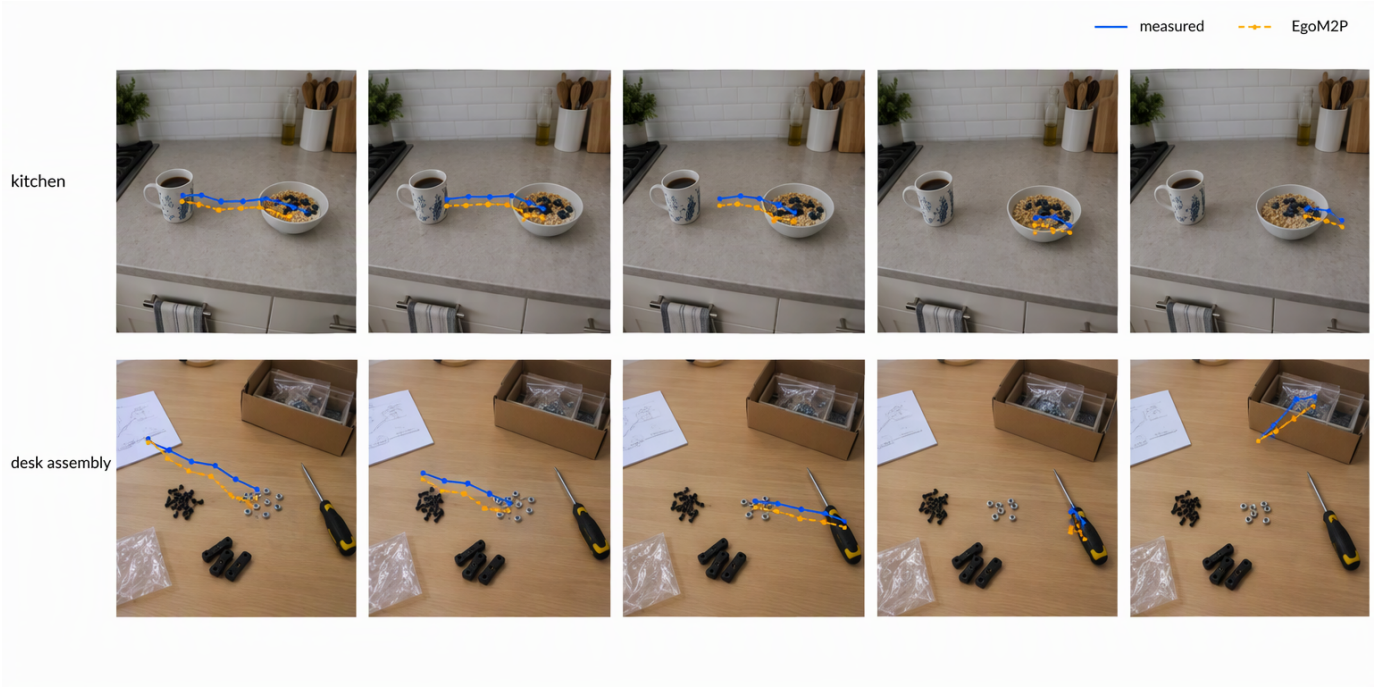



Fig. 12: EgoM2P against the L3 eye tracker in the ego view. Each row is one episode (kitchen, desk assembly) and the five columns are consecutive frames; in every frame the two short trails cover the last ten frames of gaze, solid blue for the measured record and dashed amber for EgoM2P. The estimate follows the same targets but lags at saccades and spreads during rapid head turns.